

CortexKitchen

AI-Powered Operations Intelligence for Restaurants

Full-Stack · Multi-Agent LLM System · Production-Ready Architecture

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github.com/anoushka45/cortexkitchen

A 9-node AI orchestration system that turns fragmented restaurant data into a validated, auditable operations plan — in seconds.

9

Intelligence Nodes

4

Planning Scenarios

2

Vector Collections

5

Critic Dimensions

THE PROBLEM

Restaurant managers make dozens of high-stakes calls every shift while manually piecing together demand signals, complaint history, and reservation data.

The result:

- No early warning system for high-demand nights
- Complaint data never synthesised into one view
- Inventory shortages discovered only when critical
- Zero audit trail on what was recommended and why

THE SOLUTION

CortexKitchen replaces reactive gut calls with a structured, repeatable planning workflow.

How it works:

- Manager selects a scenario – Friday rush, holiday spike, slow midweek, low stock weekend.
- agents analyse every operational signal in parallel
- Outputs merge deterministically into a unified plan
- Critic validates across 5 dimensions before delivery

AGENT ARCHITECTURE

01
Select Scenario

02
Agents Analyse

03
Outputs Merge

04
Critic Validates

05
Plan Delivered

Agents	Responsibility
Ops Manager	Classifies the scenario, sets planning parameters, short-circuits on invalid input
Demand Forecast	Synthesises historical and seasonal signals to estimate service demand with confidence bounds
Reservation Intel	Analyses booking data for occupancy risk and overbooking signals
Complaint Intel	Mines past complaints via RAG to surface recurring ops failures and relevant SOPs
Menu Intelligence	Identifies high-velocity items and scenario-specific promotion opportunities
Inventory Planner	Flags shortfalls, overstock, and spoilage risk against forecast demand
Aggregator	Merges all agent outputs into a unified planning bundle for the critic
Critic	Independently scores the plan across 5 dimensions before it reaches the operator
Final Assembler	Packages the validated plan, audit trail, and revision reasons for the dashboard

WHAT MAKES IT DIFFERENT

Structured Decision Workflow

Not a chatbot – a planning system that runs on demand and produces output a manager can immediately act on.

Critic-Validated Output

Nothing reaches the dashboard unreviewed. Five dimensions scored with specific, actionable revision reasons surfaced alongside the plan.

True Multi-Agent Orchestration

Specialised agents work in parallel across all operational signals, merging deterministically before the critic sees anything.

Audit Trail by Design

Every run persisted with critic score, cost-aware breakdown, and full agent outputs. Not an afterthought – a first-class feature.

WHAT WAS ACTUALLY BUILT

This is not a prototype or a notebook. The full deliverable includes:

- **Production-shaped monorepo:** API, frontend, infra, docs, seed data, tests
- **9 specialised intelligence nodes** with real prompts, output schemas, and error handling
- **REST API** with scenario planning, run history, and operational health endpoints
- **14-component Next.js dashboard** with live state management and drill-down modals
- **Full documentation:** architecture, data model, API contracts, evaluation rubric
- **Docker Compose infra**, Alembic migrations, vector store seed scripts

TECHNICAL DEPTH

LangGraph	Stateful DAG with typed state, reducer annotations, and conditional branching
RAG / Qdrant	2 vector collections — complaint memory, SOPs and grounded recommendations
Critic Layer	LLM scoring on 5 rubric dimensions + automated schema, quantity bounds, and 24hr feasibility checks
LLM Abstraction	Provider-agnostic: swap Gemini 2.5 Flash ↔ Groq llama-3.3-70b with one config change

ROADMAP

- **Real integrations — reservations, inventory, POS connectors**
- **Multi-tenant product layer with JWT auth and role-based access**
- **Async execution + observability pipeline**
- **Move from decision support → action-taking system**

CURRENT STATE

- End-to-end planning system live across 4 scenarios
- LangGraph orchestration + RAG-backed complaint intelligence
- Persisted runs with full audit trail and critic scores

Stack FastAPI · LangGraph · PostgreSQL · Qdrant · Prophet · Redis · Next.js · Tailwind · Docker

CortexKitchen is not another AI demo. It's a system designed to think through restaurant operations and produce decisions a manager can actually use.

Open to early pilots, collaboration, or feedback from anyone building in the ops intelligence space.